

Bariscan Kurtkaya

RESEARCH INTEREST	Exploring how memory and intelligence emerge in biological and artificial neural networks through mechanistic interpretability, neuroscience principles, and dynamical systems theory, with the goal of developing more transparent and robust AI systems.
EDUCATION	University of California, Santa Barbara, PhD in Electrical and Computer Engineering, California, USA 2025 - 2030 <i>Principal Investigator:</i> Prof. Nina Miolane Koc University, MSc in Computer Science and Engineering, Istanbul, Turkey 2025 Awarded KUIS AI Fellowship. <i>Thesis:</i> Understanding dynamical phases of short-term mechanisms with Prof. Yucel Yemez & Dr. Fatih Dinc Yildiz Technical University, BSc in Electronics and Communication Engineering, Istanbul, Turkey 2023 Graduated with High Honors.
PUBLICATIONS	* [1] <i>Dynamical phases of short-term memory mechanisms in RNNs</i>, ICML'25, 2025, [Paper] Bariscan Kurtkaya*, Fatih Dinc*, Mert Yuksekgonul, Marta Blanco-Pozo, Ege Cirakman, Mark J. Schnitzer, Yucel Yemez, Hidenori Tanaka†, Peng Yuan†, Nina Miolane† [2] <i>RAVE: Randomized Noise Shuffling for Fast and Consistent Video Editing with Diffusion Models</i>, CVPR'24 Highlight, 2024 [Paper] [Code] [Website] Ozgur Kara*, Bariscan Kurtkaya*, Hidir Yesiltepe, James M. Rehg, Pinar Yanardag† [3] <i>Learning rate collapse prevents training recurrent neural networks at scale</i>, NeurIPS'25 - NeurReps Workshop, 2025 [Paper] Bariscan Kurtkaya, Alperen Cimen*, Mehmet Harmanli*, Andy Alexander, Nina Miolane, Fatih Dinc†, Yucel Yemez† [4] <i>The Cue or not the Cue? A Mechanistic Study of Memory Mechanisms in RNNs</i>, NeurIPS'25 - NeurReps Workshop, 2025 [Paper] Mehmet Harmanli, Fatih Dinc, Peng Yuan, Yucel Yemez†, Bariscan Kurtkaya† [5] <i>Deep learning-based driver assistance system</i>, Electrica Journal, 23(3), 607-618, 2023. [Paper] [Code] Bariscan Kurtkaya, Arda Tezcan, Murat Taskiran
PREPRINTS & UNDER REVIEW	[1] <i>A ghost mechanism: An analytical model of abrupt learning</i>, Under review (Physics Journal), 2025, [Paper] Fatih Dinc*, Ege Cirakman*, Bariscan Kurtkaya*, Mert Yuksekgonul, Yiqi Jiang, Mark J. Schnitzer†, Hidenori Tanaka† [2] <i>A theory of parameter identifiability in data-constrained recurrent neural networks</i>, Under review (ML Venue), 2025 Fatih Dinc, Abby Bertics, Bariscan Kurtkaya, Nina Miolane
RESEARCH EXPERIENCE	University of California, Santa Barbara, Geometric Intelligence Lab, Graduate Student Researcher 2025 - 2030 Supervised by Prof. Nina Miolane - Exploring how memory and intelligence emerge in neural networks by developing mechanistic interpretability methods grounded in neuroscience principles and dynamical systems theory, advancing the foundation for safer and more transparent AI systems. - Analyzing memory representations and dynamics across diverse architectures (Transformers, LSTMs, GRUs) using mechanistic interpretability tools, and collaborating with experimental neuroscience labs to validate computational findings in biological neural recordings. Stanford University, Schnitzer Group, Visiting Graduate Student Researcher 2024 - 2025 Supervised by Prof. Mark J. Schnitzer & Dr. Fatih Dinc - Investigated how short-term memory mechanisms emerge in recurrent neural networks by training biologically-plausible leaky firing-rate RNNs on memory tasks. Discovered evidence for sequential activity via limit cycles that are experimentally indistinguishable from slow-point manifolds during trial periods. Developed analytical toy models and low-rank RNNs to characterize when and why different memory mechanisms emerge. Published in ICML'25. - Gained hands-on neuroscience experience by participating in mice surgeries, optogenetics experiments, and two-photon Ca^{2+} imaging sessions. Contributed to neuron segmentation work using deep learning (Cellpose), resulting in co-authorship on a manuscript.

*(with * for equal contribution, † for equal supervision)

	Koc University , Koc University AI Center , Graduate Student Researcher Supervised by Prof. Yucel Yemez - Investigated whether RNNs employ short-term versus working-memory computations in delayed discrimination tasks by developing mechanistic interpretability tools to quantify information preservation. Trained inverse layers to reconstruct input signals from hidden states across diverse training conditions, revealing that RNNs consistently demonstrate short-term memory-like computation by preserving complete input information through delay periods. Published in NeurIPS'25 NeurReps Workshop. - Identified learning-rate collapse as a fundamental barrier to training large-scale RNNs for computational neuroscience, showing that neuron count increases require proportional learning rate decreases (N^{-1}). Proposed utilizing low-rank RNN architectures to enable efficient training of networks with 100k+ neurons. Published in NeurIPS'25 NeurReps Workshop.	2023 - 2025
	Virginia Tech , GemLab VT , Graduate Research Intern Supervised by Prof. Pinar Yanardag - Developed a novel, training-free, zero-shot video editing framework, leveraging pre-trained text-to-image diffusion models to enable efficient video editing without additional training requirements. Published in CVPR'24 Highlight. - Achieved state-of-the-art performance in both qualitative and quantitative evaluations, demonstrating superior editing capabilities and addressing the shape morphing problem, a limitation of previous methods. - Curated the largest dataset to date for video editing, designed to enhance comprehensive evaluations by including diverse motion types and editing prompts, providing a robust resource for the field.	2023 - 2024
	WashU St. Louis , McDonnell Center , Undergraduate Researcher Supervised by Prof. Tansu Daylan - Conducted research to enhance exoplanet discovery through machine learning on observations from the James Webb Space Telescope. - Developed a physics-informed, GPU-accelerated data augmentation technique, overcoming data scarcity by expanding the dataset from 1000 to 10^6 in 10 minutes thereby enhancing model training capacity. - Investigated machine learning architectures and explainability methods to improve interpretability and yield scientifically meaningful insights in exoplanet discovery.	2022 - 2023
SCHOLARSHIPS	Funded by Prof. Jay McClelland through the Feldman-McClelland Open-a-Door fund. Impact Scholar, awarded a fellowship from the Bridge to Turkiye Fund. KUIS AI Fellowship, awarded fellowship from Koc University AI Center 2209 TUBITAK Research Project Grant Holder, awarded from TUBITAK.	2024 Fall 2024 Fall 2023-2025 2022
ACHIEVEMENTS	Awarded 1st, and 3rd place as in the US-based MateRov Semi-autonomous Underwater Systems competition organized under the sponsorship of NSF, in consecutive years. Awarded 4th place in the Undersecretariat of Defense Industry award in the Underwater Systems category. Awarded 3rd place in the Teknofest Turkiye award in the Underwater Systems category. Hugging Face and Microsoft Ambassador High Honors from Yildiz Technical University Top 0.25% among 150 thousand - Turkish National Postgraduate Entrance Exam and Academics Top 1% among 2 million - Turkish National University Entrance Exam	
TEACHING EXPERIENCE	ECE3 - University of California, Santa Barbara - Graduate Teaching Assistant - Delivered lab sessions on Python programming and Introduction to Data Science. Provided support during office hours, addressing student questions, and assessed student performance through grading course homeworks. COMP 410/510 - Computer Graphics - Koc University - Graduate Teaching Assistant - Delivered a lecture on 3D reconstruction using Neural Radiance Fields, introducing key concepts and applications to the class. Provided support during office hours, addressing student questions, and assessed student performance through grading course homeworks. COMP 106 - Discrete Mathematics for CSE - Koc University - Graduate Teaching Assistant - Led problem solving sessions, presenting example problems to the class on the RSA public-key cryptosystem to enhance understanding. Provided support during office hours, addressing student questions, and assessed student performance through grading midterm and final exams.	2025 Fall 2024,2025 Spring 2023 Fall
TALKS	Voxel51 CVPR2024 Meetup - Speaker - Presented our CVPR work RAVE: Randomized Noise Shuffling for Fast and Consistent Video Editing with Diffusion Models at Voxel51's CVPR meetup. Akbank MultiGroup - Women in AI - Speaker - Delivered an advanced-level seminar on deep learning-based object detection models to underrepresented groups at the Akbank MultiGroup MoreThan101 series. (Presentation Link)	2024 Summer 2023 Spring